MATH 6610: Advanced Numerical Analysis

Homework 2 — Solutions

Due: Friday, October 16, 2026

Instructions. Show all work. Numerical answers should be reported to the number of digits requested. You may use MATLAB or Python for the computational parts.

Problem 1. (Wallis's equation.) Apply Newton's method to $f(x) = x^3 - 2x - 5$ with starting value $x^0 = 2$.

(a) Compute x^1, x^2, x^3, carrying at least ten digits.

(b) The root is $x^* = 2.0945514815\ldots$ Tabulate the errors $\left|x^k - x^*\right|$ and verify that the number of correct digits approximately doubles at each step.

(c) Using the error relation $\left|x^{k+1} - x^*\right| \approx C\left|x^k - x^*\right|^2$, estimate C from your table and compare with $|f''(x^*)/(2f'(x^*))|$.

Solution. (a) With $f'(x) = 3x^2 - 2$: $f(2) = -1$, $f'(2) = 10$, so $x^1 = 2 + 0.1 = 2.1$. Then $f(2.1) = 0.061$, $f'(2.1) = 11.23$, $x^2 = 2.1 - 0.061/11.23 = 2.0945681211\ldots$; and $x^3 = 2.0945514817\ldots$

(b) Errors: $\left|x^0 - x^*\right| = 9.46 \times 10^{-2}$, $\left|x^1 - x^*\right| = 5.4 \times 10^{-3}$, $\left|x^2 - x^*\right| = 1.66 \times 10^{-5}$, $\left|x^3 - x^*\right| \approx 1.6 \times 10^{-10}$: correct digits $1 \to 2 \to 5 \to 10$, doubling as claimed.

(c) From consecutive errors, $C \approx 1.66 \times 10^{-5}/(5.4 \times 10^{-3})^2 \approx 0.57$. The theoretical constant is $|f''(x^*)|\,/(2\,|f'(x^*)|) = 6x^*/(2(3x^{*2} - 2)) = 12.567/(2 \cdot 11.16) \approx 0.563$. The agreement confirms the local quadratic model. ■

Problem 2. Consider the nonlinear system

$$x_1^2 + x_2^2 = 1, \qquad x_2 = x_1^3.$$

(a) Write the system as $f(x) = 0$ and compute the Jacobian $Df(x)$.

(b) Starting from $x^0 = (1, 1)^T$, carry out one full Newton step by hand: evaluate $f(x^0)$ and $Df(x^0)$, solve the 2×2 linear system for the correction Δx^0 exactly (fractions), and state x^1.

(c) Without further computation, explain what linear system must be solved at the next step and which parts of the work from (b) can be reused.

Solution. (a) $f(x) = \left(x_1^2 + x_2^2 - 1,\ x_2 - x_1^3\right)^T$,

$$Df(x) = \begin{bmatrix} 2x_1 & 2x_2 \\ -3x_1^2 & 1 \end{bmatrix}.$$

(b) $f(x^0) = (1, 0)^T$ and $Df(x^0) = \begin{bmatrix} 2 & 2 \\ -3 & 1 \end{bmatrix}$ with determinant 8. Solving $Df(x^0)\,\Delta x^0 = -(1, 0)^T$:

$$\Delta x^0 = -\frac{1}{8}\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/8 \\ -3/8 \end{bmatrix}, \qquad x^1 = \begin{bmatrix} 7/8 \\ 5/8 \end{bmatrix}.$$

(c) The next step solves $Df(x^1)\,\Delta x^1 = -f(x^1)$ with the Jacobian re-evaluated at x^1; nothing from the old factorization can be reused in the full Newton method. (Only the *simplified* Newton method, which freezes $Df(x^0)$, reuses the factorization; that trade-off is the subject of the next problem.) ■

Problem 3. (Simplified Newton method.) The simplified Newton iteration freezes the derivative at the starting point:

$$x^{k+1} = g(x^k), \qquad g(x) = x - \frac{f(x)}{f'(x^0)} \qquad (n = 1).$$

Let x^* be a zero of f with $f'(x^*) \neq 0$, and suppose f is continuously differentiable near x^*.

(a) Show that $g'(x^*) = 1 - f'(x^*)/f'(x^0)$.

(b) Using the mean value theorem, show that if $q := \left|1 - f'(x^*)/f'(x^0)\right| < 1$, then the iteration converges for all starting values sufficiently close to x^*, with $\left|x^{k+1} - x^*\right| \leq (q + \epsilon)\left|x^k - x^*\right|$ for any $\epsilon > 0$ eventually.

(c) Explain in one or two sentences why this method converges only linearly in general, and what one gains in exchange (think of the cost per step for systems).

Solution. (a) $g'(x) = 1 - f'(x)/f'(x^0)$; evaluating at x^* gives the claim.

(b) By continuity of g', choose $\delta > 0$ so that $|g'(\xi)| \leq q + \epsilon < 1$ for all $\xi \in [x^* - \delta, x^* + \delta]$. For x^k in this interval, the mean value theorem gives, for some ξ between x^k and x^*,

$$x^{k+1} - x^* = g(x^k) - g(x^*) = g'(\xi)(x^k - x^*),$$

using $g(x^*) = x^*$ (since $f(x^*) = 0$). Hence $\left|x^{k+1} - x^*\right| \leq (q + \epsilon)\left|x^k - x^*\right|$, the iterate stays in the interval, and by induction the errors decay geometrically to zero.

(c) Since $g'(x^*) \neq 0$ in general, the error is multiplied by an asymptotically constant factor: linear convergence. The gain for systems is that the Jacobian is factored once and each subsequent step costs only two triangular solves, $O(n^2)$ instead of $O(n^3)$; this is why the simplified correction also appears in the monotonicity test. ■

Problem 4. Compute the condition number $\kappa(x)$ of the problem $F(x) = \tan x$ using the formula $\kappa(x) = |x|\ |F'(x)|\,/\,|F(x)|$.

(a) Show that $\kappa(x) = \left|\dfrac{2x}{\sin 2x}\right|$.

(b) Determine all points in $(0, \pi)$ near which the problem is ill-conditioned, and compute $\lim_{x \to 0^+} \kappa(x)$.

(c) For $x = 1.5707$ (close to $\pi/2$), estimate $\kappa(x)$ and state how many digits of accuracy one can expect in $\tan x$ when x itself carries a relative error of 10^{-16}.

Solution. (a) $F'(x) = 1 + \tan^2 x = 1/\cos^2 x$, so

$$\kappa(x) = |x|\ \frac{1/\cos^2 x}{|\tan x|} = \frac{|x|}{|\sin x \cos x|} = \left|\frac{2x}{\sin 2x}\right|.$$

(b) $\kappa \to \infty$ exactly where $\sin 2x \to 0$ with $x \neq 0$, that is, near $x = \pi/2$ and near $x = \pi$. As $x \to 0^+$, $2x/\sin 2x \to 1$, so the problem is perfectly conditioned near the origin.

(c) At $x = 1.5707$: $2x = 3.1414$ and $\pi - 3.1414 \approx 1.93 \times 10^{-4}$, so by $\sin(\pi - t) = \sin t$, $\sin 2x \approx 1.93 \times 10^{-4}$. Hence $\kappa \approx 3.1414/1.93 \times 10^{-4} \approx 1.6 \times 10^4$. A relative input error of 10^{-16} is amplified to about 1.6×10^{-12}: roughly twelve correct digits remain, four digits lost, and no algorithm evaluating $\tan$ at this x can do better. ■

Problem 5. Consider evaluating

$$f(x) = \frac{1 - \cos x}{x^2}$$

for small x; note $\lim_{x \to 0} f(x) = \frac{1}{2}$.

(a) Evaluate the formula as written at $x = 10^{-8}$ in double precision (by hand reasoning or on a machine) and explain the result using the rounding of $\cos x$ and catastrophic cancellation.

(b) Rewrite f using the identity $1 - \cos x = 2\sin^2(x/2)$ and show that the new formula evaluates accurately for small x.

(c) Which of the two computations is unstable, and is the underlying *problem* ill-conditioned at small x? Justify with the condition number of f near 0.

Solution. (a) For $x = 10^{-8}$, $\cos x = 1 - 5 \times 10^{-17} + \cdots$, which rounds to exactly 1 in double precision ($5 \times 10^{-17} < \frac{1}{2}\varepsilon_{\text{mach}}$ relative to 1). The numerator evaluates to $1 \ominus 1 = 0$, so the computed f is 0 instead of ≈ 0.5: total loss of accuracy. The subtraction cancels all correct digits and promotes the representation error of $\cos x$ to relative size 1.

(b) With $f(x) = \dfrac{2\sin^2(x/2)}{x^2} = \dfrac{1}{2}\Big(\dfrac{\sin(x/2)}{x/2}\Big)^2$, the quantity $\sin(x/2) \approx 5 \times 10^{-9}$ is computed with relative error $O(\varepsilon_{\text{mach}})$, and the formula involves only multiplication and division of well-scaled quantities: the result is 0.5 to nearly full precision. No subtraction of close numbers occurs.

(c) The naive computation is unstable; the problem itself is well conditioned near 0: writing $f(x) = \frac{1}{2} - \frac{x^2}{24} + \cdots$, we get $\kappa(x) = |xf'(x)/f(x)| \approx |x \cdot (-x/12)/(1/2)| = x^2/6 \to 0$. Small input errors are damped, so an accurate algorithm is possible, and (b) exhibits one. The failure in (a) is entirely the algorithm's. ■